

OSSP (Operating Systems And Systems Programming)

Course code: 25CS2104E

Practical-01: Linux Command Execution

Code output:

```
sushma_sree@LAPTOP-6PJSRT08:~$ nano practical1.c
sushma_sree@LAPTOP-6PJSRT08:~$ gcc practical1.c
sushma_sree@LAPTOP-6PJSRT08:~$ ./a.out
Enter Linux Command:ls
Parent Process
Parent PID: 941
Child Process
Child PID: 942
Parent PID: 941
a.out  a9      b2      execeexample2.c  f6      linux_task4  pipe10.c  pipes1.c  sush.c.save  z7
a2     a9.save  dir1    execeexample3.c  f6.c    mvex.c      pipeex1.c  practical1.c  sush.c.save.1  z8
a3     abc     dir3    execeexample.c   fork4.c  open1.c     pipeex1.c  readwr.c     sush.c.save.2
a4     ass1.c  dirz    f1               linux_task1  open3.c    pipeex2.c  student_portal  task1.c
a6     b1      ex1     f1.c            linux_task2  permission_lab  pipes.c    sush.c      xyz
Child Process Completed
sushma_sree@LAPTOP-6PJSRT08:~$ |
```

B) Linux terminal commands: (uname, lscpu, lsblk, ps, top)

uname:

```
sushma_sree@LAPTOP-6PJSRT08:~$ uname
Linux
sushma_sree@LAPTOP-6PJSRT08:~$ |
```

lscpu:

```
sushma_sree@LAPTOP-6PJSRT08:~$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:           39 bits physical, 48 bits virtual
Byte Order:              Little Endian
CPU(s):                  8
On-line CPU(s) list:     0-7
Vendor ID:               GenuineIntel
Model name:              12th Gen Intel(R) Core(TM) i3-1215U
CPU family:              6
Model:                   154
Thread(s) per core:      2
Core(s) per socket:      4
Socket(s):               1
Stepping:                4
BogoMIPS:                4992.00
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopt xtopology tsc_reliable
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
1 xsave avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
ear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:          VT-x
Hypervisor vendor:       Microsoft
Virtualization type:     full
```

```

Caches (sum of all):
  L1d: 192 KiB (4 instances)
  L1i: 128 KiB (4 instances)
  L2: 5 MiB (4 instances)
  L3: 10 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-7
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_D
               IS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
sushma_sree@LAPTOP-6PJSRT08:~$

```

lsblk:

```

sushma_sree@LAPTOP-6PJSRT08:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    1G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
sushma_sree@LAPTOP-6PJSRT08:~$

```

ps:

```

sushma_sree@LAPTOP-6PJSRT08:~$ ps
  PID TTY          TIME CMD
   805 pts/0        00:00:00 bash
  1068 pts/0        00:00:00 ps
sushma_sree@LAPTOP-6PJSRT08:~$

```

top:

```
sushma_sree@LAPTOP-6PJSRT08:~$ top
top - 11:15:06 up 1:03, 1 user, load average: 0.02, 0.04, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3779.6 total, 3161.1 free, 433.7 used, 259.5 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3346.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1094	sushma_+	20	0	8152	6040	3900	R	0.3	0.2	0:00.07	top
1	root	20	0	24168	15416	11604	S	0.0	0.4	0:01.36	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.1	0:00.02	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.1	0:00.00	init
46	root	19	-1	50400	16752	15464	S	0.0	0.4	0:00.58	systemd-journal
81	systemd+	20	0	22416	14488	11988	S	0.0	0.4	0:00.31	systemd-resolve
90	root	20	0	35264	12212	9232	S	0.0	0.3	0:02.34	systemd-udevd
110	root	20	0	2888	1948	1820	S	0.0	0.1	0:00.06	chronyd-starter
111	root	20	0	4472	3128	2880	S	0.0	0.1	0:00.02	cron
112	message+	20	0	8792	5352	4676	S	0.0	0.1	0:00.09	dbus-daemon
141	root	20	0	44096	29740	14600	S	0.0	0.8	0:00.37	networkd-dispat
156	root	20	0	18440	9472	8224	S	0.0	0.2	0:00.19	systemd-logind
182	syslog	20	0	220548	5948	4660	S	0.0	0.2	0:00.14	rsyslogd
191	root	20	0	5492	2728	2544	S	0.0	0.1	0:00.00	agetty
200	root	20	0	123460	32560	17952	S	0.0	0.8	0:00.33	unattended-upgr
229	_chrony	20	0	20424	7084	6292	S	0.0	0.2	0:00.29	chronyd
239	_chrony	20	0	12096	2444	1804	S	0.0	0.1	0:00.00	chronyd
324	root	20	0	8644	5592	4752	S	0.0	0.1	0:00.04	login
374	sushma_+	20	0	22336	13588	10972	S	0.0	0.4	0:00.24	systemd
378	sushma_+	20	0	22912	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
404	sushma_+	20	0	6136	5512	3928	S	0.0	0.1	0:00.04	bash
799	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
800	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.08	Relay(805)
805	sushma_+	20	0	6268	5548	3872	S	0.0	0.1	0:00.35	bash